Actual first-jet factors with additive moment weights

43 authored statements, 14 November 2026
Source `6e97fcc1fffc1dcbdc25770aac701c2e252905c5`

Result scope. Constructed integral factors for the actual completed first crossing jet, with $2^u k$ labels, normalized residual k, additive weighted row profile $w + b$ and absolute residual $w + k$. These are unsigned source-bound component claims [pcc:unverified].

The same actual polynomial. Keep the orientation-completed family $W_{xy}(T)$, actual retained bit mask

$$M_{xy} = \{j < u : \mathrm{bit}_j(x) = \mathrm{bit}_j(y)\}, \qquad \delta(x, y) = y - x.$$

At $t = -2^i$, $i < u \le a$, let J_{xy} be coefficient one of $W_{xy}(t + 2^{a+1}Z)$. The parent derives the actual nonzero diagonal $c = J_{00}$ and $w = v_2(c) = a + 1 - i + G_i$. Let $R_{xy} = J_{xy}/c$ at equal-bit-i crossings and zero elsewhere. All source ports are in $\mathrm{Fin}(2^a)$; retained multiplicities are positive. Nodes, origin and source degree stay those of this same completed family.

Actual row coordinate. Set

$$r = x \bmod 2^u, \qquad h = x \operatorname{div} 2^u, \qquad x = r + 2^u h.$$

These are literal integer operations. The actual masks at (x, y) and (r, y) agree. For the pole $q = t$ and each selected anchor $q = 2^j$, define

$$d_q = q - \delta(r, y), \qquad e_q = 2^u/d_q.$$

Every relevant denominator is nonzero. Its actual valuation is at most $i < u$ for the pole or $j < u$ for a selected anchor, hence $v_2(e_q) \ge 1$. The literal row identity gives

$$\frac{d_q}{q - \delta(x, y)} = (1 + e_q h)^{-1}.$$

No favorable denominator or scalar is supplied.

Actual reciprocal family. Polynomial derivative/first-jet identities and exact finite products give

$$R_{xy} = R_{ry}/D_{ry}(h), \qquad D_{ry}(H) = (1 + e_t H) \prod_{j \in M_{ry}} (1 + e_j H)^{m_j}.$$

The constructed finite polynomial depends on the retained row class r, y, not its quotient h. Its constant is one. Convolution, finite products and powers of the actual even slopes prove

$$\mathrm{VBound}(b, [H^b]D_{ry}), \quad \mathrm{VBound}(s, q) \equiv (q = 0 \vee v_2(q) \ge s).$$

These are actual-family identities; no approximation oracle or desired coefficient bound is assumed. PCC-20261114-84a6cb [pcc:unverified].

Constructed inverse, truncation and finite error

For the actual weighted denominator D with $D_0 = 1$, construct $F = \mathrm{invOfUnit}(D, 1)$ with the pinned standard power-series API. Its explicit coefficient recurrence is

$$f_0 = 1, \qquad f_n = -\sum_{a+b=n,\ b<n} D_a f_b \quad (n > 0).$$

Strong induction proves $\mathrm{VBound}(n, f_n)$: every summand has additive weight $a + b = n$, and finite sums preserve the bound. The actual constant-one identity gives $DF = 1$. This is a constructed inverse, not an assumed good series. Standard inverse-series arithmetic is reused.

Finite approximation. For any $k \geq 0$, define

$$Q_k(H) = \sum_{b<k} f_b H^b.$$

Its coefficients retain their degree weights. The finite polynomial $E = DQ_k - 1$ has zero coefficients below k by the exact formal inverse identity; every coefficient E_b retains valuation at least b. For any actual integer h, each surviving term $E_b h^b$ has valuation at least k, so

$$\mathrm{VBound}(k, E(h)).$$

Only the finite polynomial Q_k is evaluated. This error follows algebraically from the finite residual $DQ_k - 1$.

Actual denominator unit. The same weighted-coefficient argument applied to $D - 1$ gives $D(h) \equiv 1 \pmod 2$. Consequently $D(h) \neq 0$ and $v_2(D(h)) = 0$. Dividing the finite residual derives

$$\mathrm{VBound}\big(k, D(h)^{-1} - Q_k(h)\big).$$

All zero exclusions and unit valuation are derived. PCC-20261114-a88b83 [pcc:unverified].

Multiplying by the actual integral prototype R_{ry} and using the exact reciprocal-family identity gives

$$\mathrm{VBound}\big(k, R_{xy} - R_{ry} Q_k(h)\big).$$

Prototype integrality comes from the parent's actual first-jet proof. The row-class invariance of D is established literally, so its coefficients can be indexed by r, y in the finite factors on the next page. Noncrossing prototype entries are explicitly zero.

The proof retains the exact hypotheses on the original bit mask, finite ports, positive multiplicities and retained level. It assumes neither the desired matrix factorization nor a favorable truncation. Coefficient construction and rational bit lengths still need complete cost accounting; a valuation estimate alone does not price them.

Explicit finite factors and actual weights

Labels are actual pairs $(r, b) \in \mathrm{Fin}(2^u) \times \mathrm{Fin}(k)$. Writing $f_b(r, y)$ for the constructed inverse coefficient, define

$$A_{x,(r,b)} = [x \bmod 2^u = r], 2^b (x \operatorname{div} 2^u)^b,$$
$$B_{(r,b),y} = R_{ry}\, f_b(r, y)/2^b.$$

If r, y is noncrossing then $B = 0$. Otherwise the actual prototype is integral, and the derived coefficient weight b cancels the exact valuation of 2^b. Thus

$$\mathrm{VBound}(b, A_{x,(r,b)}), \qquad \mathrm{VBound}(0, B_{(r,b),y}).$$

No favorable integral factor is supplied. PCC-20261114-3be17e [pcc:unverified].

Exact product. The class selector leaves one literal residue $r = x \bmod 2^u$. The factors 2^b cancel, and the finite sum over $b < k$ is exactly the constructed polynomial evaluation. Therefore

$$(AB)_{xy} = R_{ry} Q_k(h), \qquad \mathrm{VBound}(k, R_{xy} - (AB)_{xy}).$$

All crossings and noncrossing zeros are included. PCC-20261114-15d101 [pcc:unverified].

Absolute weights. Multiply the actual row factor by the derived c. The exact matrix-product scaling gives

$$\mathrm{VBound}(w + b, (cA)_{x,(r,b)}), \qquad \mathrm{VBound}(w + k, (J_{\mathrm{cross}} - cAB)_{xy}).$$

Feasibility $G_i \geq 0$ makes cA integral as well. The weights are additive: $w + b$, without a product of the gain and moment degree. PCC-20261114-e3550d/f04bf6 [pcc:unverified].

Charged count. The exact number of inner labels is

$$\#\big(\mathrm{Fin}(2^u) \times \mathrm{Fin}(k)\big) = 2^u k.$$

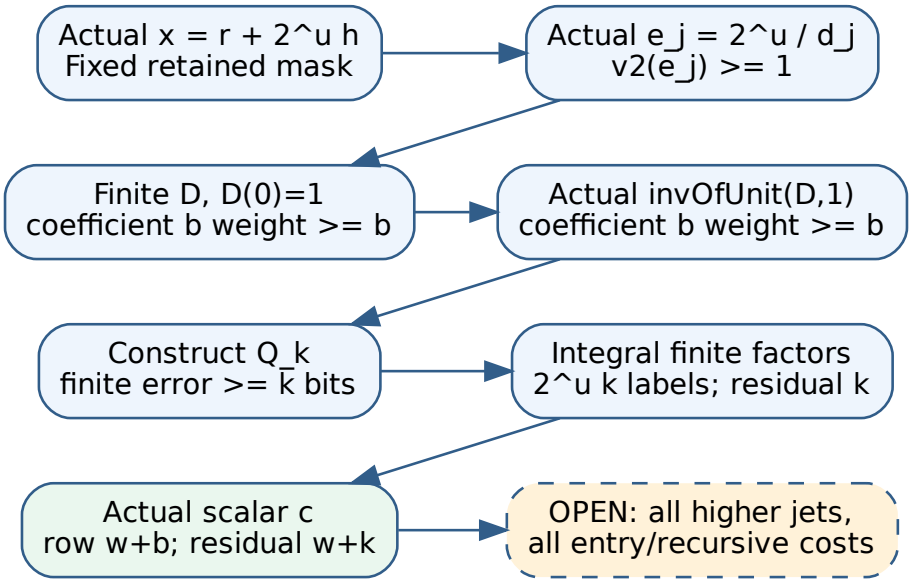
For $k = 0$ the factors are empty and the residual bound is zero bits. For $k \geq 1$, $k \leq 2^{k-1}$ gives

$$2^u k \leq 2^{u+k-1},$$

the earlier row-refinement upper count when $u + k - 1 \leq a$. This comparison is between symbolic label counts. It is not a timing or complete bit-cost comparison with the prior source's existing linear-moment method. At $k = 1$ it matches the independently proved necessary count 2^u for this actual first-jet precision. PCC-20261114-f6a09d [pcc:unverified].

Every retained mask is charged. The factor 2^u can exceed the old 2^{i+1} tile count. All required higher chart jets and the full source mode/precision/recovery/entry/recursion interface remain separate obligations.

Evidence and remaining algorithmic obligations



Formal artifacts. One module contains43 authored statements. Pinned shared gate audits44 declarations, including generated `momentRight.congr_simp`, with0 extra trusted axioms, exit0. Import-inclusive fresh OrientationMoment exits0 with default imports. All exact source-bound records are unsigned; gate data PCC-20261114-24441c [pcc:unverified].

Separate higher-chart draft. Seven isolated statements locally derive actual retained crossing base content $a+1-i$, whole completed crossing chart content w and coefficientwise integrality after division by the actual c. They are outside this43/44 audit/fresh, unsigned data310194. They provide no higher-chart factorization [pcc:unverified].

Evidence preservation. Source/dependency pins, all failed drafts and exact raw logs are retained. No old source, claim body or canonical ledger history is rewritten. Parent fixed literal scaling certificate checks its stated ring rearrangement only; no new native crypto fixture, search, performance panel, paid/cloud campaign or old native tests were run.

Prior work and next discrimination. KB unavailable; bounded fallback and primary readings recorded. Actual prior LocalCharts inverse construction, repeated-anchor linear moments and pinned Mathlib inverse/truncation proofs are reused without a priority claim. Next prove the required higher-chart factor profile and price the actual weighted mode density, coefficient/bit/ entry/recovery/recursion work. Efficient public-input finder/extraction and complete expected ECDLP bit cost remain needed. Origin remains identity. PR4118 protected-history reconciliation is a separate integration dependency.

`moments/PROOF_MAP.md`
`moments/evidence/claim-pins.json`
`moments/evidence/gate-validation-001.json`